2016 EXAMINATIONS

Lancaster University

Part I (First Year)

SCHOOL OF COMPUTING AND COMMUNICATIONS

SCC.160 Fundamentals of Communication Systems (1 hour & 30 Minutes)

- **Answer any THREE out of the four questions.**
- **Use a separate answer book for each question.**

Question 1

1.a Explain the following two concepts:

i). Digital signals

[2 marks]

ii). Analogue signals

[2 marks]

1.b Recall that, to express a quantity Y in dB (decibels), we use the formula $10\times\log_{10}(Y)$. We also know that $\log_{10}(a\times b)=\log_{10}(a)+\log_{10}(b)$, $\log_{10}(2)\approx 0.3$ and $\log_{10}(10)=1$. Answer the following questions:

i). The background noise in a busy supermarket is about 4,000 times higher than the hearing threshold. What is the noise level in dB?

[5 marks]

ii). The sound level of the subway is about 19 dB. How many times is the subway sound higher than the hearing threshold?

[5 marks]

1.c Describe the following two modulation types:

i). Amplitude shift keying (ASK)

[3 marks]

ii). Frequency shift keying (FSK)

[3 marks]

1.d Calculate the symbol rates for the given bit rates and types of modulation:

Bit Rate	Modulation Type	Symbol Rate
2000 bps	16-PSK	
6 kbps	8-QAM	
6000 bps	FSK	
35 kbps	128-FSK	
128 kbps	16-PSK	

[5 marks]

[Total 25 Marks]

Question 2

2.a Explain the differences between *low level*, *middle level* and *high level* image processing tasks.

[6 marks]

2.b What are the main components of digital camcorders? Explain the camcorders' principle of work. What is the operation of each component?

[5 marks]

2.c i). What is the intensity histogram of an image?

[2 marks]

ii). Calculate the *intensity histogram* for the grey-scale image below.

7	2	1
4	4	3
5	0	1
5	5	1
7	7	5

[4 marks]

2.d Explain the very basic principle of automatic detection of objects in video sequences, known as *the frame differencing approach*. Describe at least *six* problems that this approach entails.

[6 marks]

2.e Why do we segment objects in images? Give *four* reasons.

[2 marks]

[Total 25 Marks]

Question 3

3.a Define the grades of Audio/Speech signal quality with respect to the input signal bandwidth. Which types of speech sounds, i.e. voiced and/or unvoiced, are affected by passing them through a low pass filter whose cut-off frequency is 3.4 KHz and why?

[6 marks]

3.b Explain what happens if the sampling frequency is:

i). bigger than twice the signal bandwidth
ii). equal to twice the signal bandwidth
iii). less than twice the signal bandwidth

[6 marks]

3.c Provide the general block diagram of:

i). A Speech Synthesis Process that is based on the "Binary Excitation Model".

[4 marks]

ii). A Speech Synthesis Process that is based on a "Mixed Excitation model".

[4 marks]

3.d Obtain the output bit rate of a pulse-code modulation (PCM) encoder, which samples a voice signal occupying a bandwidth of 7 kHz with Nyquist sampling rate, if each sample is coded by 8 bits.

[5 marks]

[Total 25 Marks]

Question 4

4.a i). Alice transmits encrypted messages to Bob, while Eve tries to intercept them. Draw a block diagram of a cipher system using the correct terminology.

[4 marks]

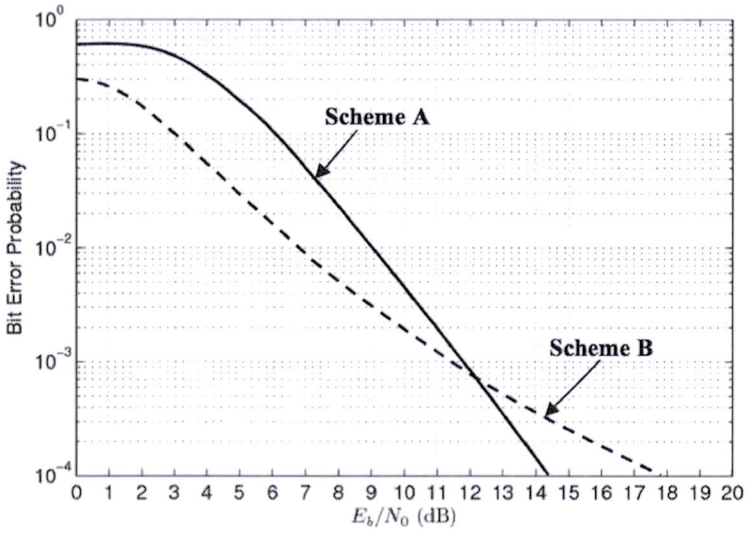
ii). List the *three* worst-case scenarios that are usually assumed when assessing the security of a system.

[3 marks]

4.b The Inmarsat system uses four geostationary satellites to provide its mobile satellite service. Describe two advantages and two disadvantages of using geostationary satellites for a satellite telephone system.

[5 marks]

4.c The bit error probability of transmission schemes A and B is depicted in the figure below. Both schemes have the same code rate and the same modulation order.



Answer the following questions:

i). Which scheme would you recommend for E_b/N_0 values higher than 13 dB and why?

[2 marks]

ii). Which scheme would you recommend for a service that requires a bit error probability of 4×10^{-4} and why?

[2 marks]

iii). If 6000 bits are transmitted using scheme B at E_b/N_0 = 10 dB, how many bits should we expect to receive in error?

[2 marks]

Question 4 continues on the next page

Question 4 continued.

4.d i). Describe the components and operation of transmitting antennas used in the ground segment of satellite TV systems.

[3 marks]

ii). Briefly explain what antenna gain is and how it is affected by the size of the antenna, its operating frequency and its beam-width.

[4 marks]

[Total 25 Marks]

END OF PAPER